DSHFNet: Dynamic Scale Hierarchical Fusion Network Based on Multiattention for Hyperspectral Image and LiDAR Data Classification

Yining Feng, Liyang Song, Lu Wang, and Xianghai Wang

***Abstract*— With the continuous improvement of satellite sensor performance, it is becoming easier to obtain different types of remote sensing (RS) data from multiple sensors, and the fusion of hyperspectral (HS) images and light detection and ranging (LiDAR) for land use/land cover (LULC) classification has become a research hotspot. However, the current mainstream methods still have defects in feature extraction and feature fusion. In the feature extraction stage, previous methods usually use a single-scale patch as input and a fixed convolution kernel for feature extraction, which makes it difficult to extract features in line with different land cover types at the same time and to obtain high-quality features. Although multiscale feature extraction can solve the one-sidedness problem of single-scale features, it also brings the challenge of high-dimensional multiscale features. In the feature fusion stage, the current fusion methods are relatively simple. Therefore, we propose a dynamic scale hierarchical fusion network (DSHFNet) for fusion classification of HS images and LiDAR data. By calculating the similarity in the scale space and judging the information at different scales through the threshold value, the appropriate scale features are dynamically selected, the small-scale features are integrated into the large-scale features, and the dimensionality of the features is reduced. This method solves the unreliability problem of single-scale features and the high-dimensional problem of multiscale features. In the feature fusion process, different attention modules are used for hierarchical fusion, spatial attention modules are used for shallow fusion and joint feature extraction, and modal attention modules are used for deep fusion of joint features and features from different sensors to achieve complete complementarity of features. Experimental evaluations on three real RS datasets demonstrate the superiority of the proposed method compared with existing methods. The source code can be downloaded at https://github.com/SYFYN0317/DSHFNet.**

***Index Terms*— Attention mechanism (AM), deep learning, hierarchical fusion, hyperspectral (HS)-light detection and ranging (LiDAR) classification, multiscale feature.**

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Yining Feng is with the School of Geography, Liaoning Normal University, Dalian 116029, China (e-mail: syfyn@outlook.com).

Liyang Song and Lu Wang are with the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: 3273687973@qq.com; wanglu2000a@163.com).

Xianghai Wang is with the School of Geography and the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: xhwang@lnnu.edu.cn).

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I. Introduction

IN RECENT years, more multisource remote sensing (RS) information can be obtained by relying on satellite-based and unmanned aerial vehicle (UAV)-based Earth observation platforms. The development of multisource cooperative Earth observation technology makes it possible for land use/land cover (LULC) to obtain multidimensional information, thus laying a foundation for the research on land cover interpretation [1], [2], [3], [4]. Hyperspectral (HS) RS has a wide spectral range, high spectral resolution, and the advantage of combining imagery and spectrum, but lacks effective elevation information [5]. Given the ability of light detection and ranging (LiDAR) data to capture various types of elevation information, and effectively compensate for the absence of elevation information in HS image, fusion HS image with LiDAR data has emerged as a key research area in the current LULC research landscape [6].

The feature-level fusion method is more suitable for the fusion of multisource heterogeneous RS data [7]. Feature-level fusion classification aims to extract and collect features of multisource RS data to form feature descriptions with richer content and higher identification, thus laying a foundation for classification decisions. The traditional feature level fusion methods include simple feature combination methods based on serial or parallel, feature selection methods based on genetic algorithm, and feature transformation methods based on D-S evidence theory and fuzzy logic [8]. Rasti et al. [9] extracted unique features from different sensors and fused spectral, spatial, and elevation features with total variation component analysis (OTVCA). Liao et al. [10] proposed a fusion method based on the generalized graph to carry out coupling reduction and feature fusion. Samiappan et al. [11] proposed a fusion method based on morphology and random feature selection algorithm and used elevation information of LiDAR data to solve the problem of data complexity. Traditional methods more rely on researchers to design feature extraction methods. How to extract the most effective features and how to conduct object labeling still need further exploration.

In recent times, the fusion classification technique using deep neural networks (DNNs) for HS image and LiDAR has undergone significant progress [12]. Xu et al. [13]

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proposed a two-branch convolutional neural network (TBCNN) and introduced spectral domain branches and spatial domain branches to conduct a collaborative classification of HS images. Hang et al. [14] designed two coupled convolutional neural networks (CoupledCNNs) to learn the spatial–spectral features of HS images and the elevation features of LiDAR data, and finally realized the coupling of the two convolutional layers by parameter sharing. Roy et al. [15] proposed a joint feature learning and fusion by CNNs and morphological blocks to extract LiDAR features using morphological expansion and erosion layers. Zhao et al. [16] proposed a fractional Gabor convolutional network (FGCN), which emphasizes both effective integration of features and thorough extraction of relevant features. Wu et al. [17] proposed a cross-channel reconstructed convolutional neural network (CCR-Net) to learn a more compact fusion representation through cross-modal reconstruction strategies, which can exchange information with each other in a more efficient manner. Zhang et al. [18] proposed an interleaving perception CNN (IP-CNN) to improve the joint classification performance. Gao et al. [19] embedded adversarial complementary learning (ACL) strategies into the CNNs (ACL-CNNs) to extract complementary information from multisource data. Zhang et al. [20] proposed a structural optimization transmission network (SOT-Net) for strengthening the semantic relatedness of multisource. Apart from CNNs, other advanced neural network structures have also been used by researchers to research, such as autoencoders (SAEs) [21], graph convolutional networks (GCNs) [22], generative adversarial networks (GANs) [23], Transformers [24], [25], [26], [27], [28], etc. All the results have yielded positive outcomes. Among them, the Transformer framework has become very popular in recent years. He et al. [24] proposed a classification architecture using the bidirectional encoder representation from transformers (HSI-BERT), which could capture the global dependence between pixels. Zhao et al. [25] proposed fractional Fourier image transformer (FrIT) as the backbone network, which can greatly accelerate the speed of transformer architecture and effectively learn valuable contextual information. Zhang et al. [26] proposed a new model called multimodal transformer network (MTNet), which uses the advantages of transformers to capture specific and shared features. Yao et al. [27] proposed a novel multimodal deep learning framework by extending conventional visual transformer (ViT) with minimal modifications (ExViT) aiming at the task of land use and land cover classification. Ding et al. [28] proposed global–local transformer network (Glt-Net) that fully utilizes CNNs' advantages in characterizing local features and Transformer's advantages in learning long-distance dependence. Although Transformers can better extract the long-distance dependence between images, it is not perfect. Its parallel processing mechanism is at the cost of inefficient reasoning and occupies a large memory, which is not suitable for directly applying to the multisource RS image fusion classification task with a large amount of data.

Although there have been several innovative methods for HS images and LiDAR fusion and classification, there are still limitations in achieving sufficient feature extraction. These limitations can be summarized as follows.

1) CNNs, as a mainstream backbone architecture, have shown strong capabilities in extracting spatial structure information and local contextual information from HS images. However, CNNs often use single-scale patches as inputs and use fixed convolutional kernels for feature extraction. Different land cover types are influenced by different surrounding distributions and require varying input scales. Using a fixed scale as input makes it difficult to simultaneously meet the practical needs of different land cover types and cannot achieve fine-grained land cover classification.
2) Multiscale input can extract features of land cover from different scales, which can form complementary joint features and improve classification accuracy. However, the input of multiscale data presents two challenges. On one hand, the contribution of different scales to classification performance is not equal, and thus, it is necessary to calibrate the corresponding weights of each scale. On the other hand, simply concatenating multiscale features may exacerbate the dimensionality problem, especially when there is a lack of labeled data, which could hinder the final classification performance.

Consequently, we propose a dynamic scale hierarchical fusion network (DSHFNet) that overcomes several challenges. The network dynamically extracts the most suitable scale features, effectively overcoming the limitations of traditional methods that may fall short in feature extraction. Simultaneously, fusing small-scale features into large-scale features helps reduce feature dimensions. Considering the interdependencies among features from different sensors, using a spatial attention module for shallow fusion and joint feature extraction, a modality attention module for deep fusion of joint features and features from different sensors, and feature weights obtained during the feature extraction process to generate more robust features, improves the classification accuracy. The primary innovations of this article are enumerated as follows.

1) A DSHFNet is proposed for HS image and LiDAR classification task, which leverages relevant features from multiple scales to generate joint features with stronger complementary advantages and significantly improves the classification accuracy.
2) A dynamic scale feature extraction (DSFE) module has been developed to address the challenge of insufficient feature extraction in traditional methods. Different from the existing multiscale feature extraction methods, aiming at the common problem of insufficient representation of single-scale features and excessive dimension of multiscale features, we calculate and evaluate the similarity of features in scale space, dynamically select appropriate scale features, integrate small-scale features into large-scale features, and reduce the dimension of features, thus effectively solving these two problems.
3) Hierarchical fusion mechanism for feature fusion is used in the proposed method, which uses different attention modules for feature fusion. Different from the existing

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methods, which only consider the relationship between shallow features and deep features, the unique features and joint features of the model are not effectively combined. The spatial attention module is deployed for shallow feature fusion and joint feature extraction, and the modal attention module is used for deep fusion of joint features and different sensor features. Through the interaction of multisource features and joint features, sufficient hierarchical fusion of features is achieved.

4) Three publicly available datasets are used to evaluate the proposed method, and some state-of-the-art (SOTA) HS image and LiDAR classification algorithms are compared with it. The findings from the experimentation indicate that DSHFNet exhibits exceptional performance, with an accuracy of 92.88% on the Houston 2013 dataset, at least 2.5% higher than other advanced algorithms.

The remainder of this article is structured as follows. Section II provides a summary of the existing literature on multiscale feature extraction, hierarchical fusion, and attention mechanisms (AMs). In Section III, the proposed framework and the novel modules are introduced in detail. Section IV presents the experimental setup, results analysis, and ablation experiments. Finally, Section V provides the article's conclusion.

II. Related Work

A. Multiscale Feature Extraction

As the basic step of most computer vision and multimedia processing tasks, feature extraction and representation have been widely studied by researchers, especially in deep learning. The ability to extract high-quality features is particularly crucial for efficient parameter training in subsequent networks [29]. Using different scales to extract features, different information can be observed and different tasks can be completed. Smaller scales can obtain more detailed local information, and larger scales can get the overall global trend. The description of image features established by multiscale features can extract more comprehensive information and obtain better results [30].

In RS image processing, there are often widespread problems such as spectral similarity, interleaved edges of complex scenes, and multiple singular points among mixed vegetation among land cover categories. Hence, the efficient extraction of significant land cover information attributes is crucial in determining the interpretational results accuracy. At present, there are many researches on RS image processing and analysis using multiscale features. Gao et al. [31] proposed a new multiscale spectral spatial cross extraction network (MSCEN) for HS image classification. Pan et al. [32] investigated HS image denoising using multiscale adaptive fusion networks (MAFNets). Wang et al. [33] proposed MashFormer, an innovative multiscale sensing integrated hybrid detector with CNNs and Transformer to enhance the characterization capability in complex background scenes. It can improve the detection performance of targets with multiscale features, so as to complete the target detection task with better precision.

In recent years, some research studies on collaborative classification based on multiscale have attracted much attention. Xue et al. [34] proposed a new multiscale DNNs, which uses hierarchical residual structure combined with self-calibrated convolution to extract features of different receiving domains and enhances the model's ability to represent multimodal data. Ding et al. [28] proposed a novel Glt-Net, which extracts multiscale local spatial features and performs an adaptive linear weighted fusion of multimodal features. Meanwhile, spectral features of HS data are integrated with multiscale features. Song et al. [35] proposed a multiscale pseudo-Siamese network with AM (MA-PSNet) and introduced a multiscale feature learning module to fully extract features at different scales.

Although multiscale feature input compensates for the information limitations of single scale features, it is still very challenging. On one hand, different land cover types have different requirements for different scales, that is, the contribution of multiscale features to classification performance is not equal. Therefore, it is necessary to calibrate the corresponding weights of each feature, which is a laborious and standard uncertain process. On the other hand, simply combining multimodal features from different scales through cascading may exacerbate high-dimensional problems, while high-dimensional data may lead to the challenge of dimension curse in the model when labeled data are insufficient, which may ultimately affect the accuracy of the final classification of multimodal data. Therefore, selecting appropriate scale features among multiscale features and effectively using multiscale feature information to avoid excessively high dimensionality of features are key issues that urgently need to be addressed.

B. Hierarchical Fusion

The traditional single-layer fusion methods face difficulty in simultaneously keeping texture information in shallow features and the semantic information in deep features. Researchers have been attempting to address this issue using multimodal DNN, but its robust learning representation ability is still limited in the face of complex scenes with mixed attributes of different ground objects. This is particularly true when there are only a insufficient number of training marker samples available and the quality of the markers themselves is subpar. This single-level feature fusion method limits the effective correlation of multisource information to a large extent, thus weakening the complementary support between multisource RS data features. Hierarchical fusion effectively reduces the spatial size of cascading features, thereby reducing dimension disasters and additional computational complexity costs. Tu et al. [36] proposed a global–local hierarchical weighted fusion end-to-end framework for HS image classification. Zhang et al. [37] designed a hierarchical dynamic fusion network (HDFNet) to detect the changes in optical aerial images.

The fusion classification of HS image and LiDAR using hierarchical fusion has also garnered significant interest among researchers. Zhao et al. [38] used an effective hierarchical random walk network (HRWN) to study the joint classification of HS images and LiDAR data, and jointly optimized the dual

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tunnel CNN and pixel affinity through a novel Random walk layer, thereby enhancing the deeper spatial uniformity of the network. Zhang et al. [39] proposed an unsupervised feature extraction framework based on patch-to-patch convolutional neural network (PToPCNN), in which the hierarchical fusion module can make optimal use of the joint features extracted by PToPCNN for classification. Peng et al. [40] developed a multiview hierarchical network (MVHN) for HS image and LiDAR data classification. Wang et al. [41] proposed a multiattentive hierarchical dense fusion net (MAHiDFNet) for the fusion classification of HS image and LiDAR data, which improves the effectiveness of fusion features through a spatial–spectral elevation three-branch framework combined with an attention module. Xue et al. [42] developed a new depth-level vision transformer (DHViT) for joint classification. Most existing hierarchical fusion strategies focus on extracting more comprehensive joint features but lack research on the interaction of features from multiple sensors.

C. Attention Mechanism

AM has been a research topic in neuroscience for several decades. The AM originated from the human visual system. When human visual system observes things in the outside world, the observed things are generally not noticed as a whole but usually selectively acquire important parts according to needs, and then form a whole impression on the observed things by combining the information of these different parts. That human vision does not process the whole thing at once, it just focuses. The AM is of utmost importance as it is responsible for enhancing the performance and precision of DNNs [43]. The AM can assign different weights to the concerned parts of each input by the model so that the model can focus on extracting the more critical and important content of the given information, so as to make more accurate judgments. Meanwhile, the use of AM does not bring more overhead to the model storage and computation, which is one of the important reasons that AM is so widely used [44].

The AM has been widely applied in visual tasks by enhancing the features. Hu et al. [45] proposed a squeeze-and-excitation network (SENet), which emphasizes features that are correlated and suppress those that are weakly or irrelevantly related. Woo et al. [46] proposed a convolutional block attention module (CBAM) that can enhance features in both the channel and spatial domains. Dosovitskiy et al. [47] presented a ViT based on the traditional transformer model and applied self-AM to extract image features, achieving results comparable to those of CNNs.

The success of AM in computer vision has introduced new concepts to the field of RS image processing. Zhu et al. [48] proposed an end-to-end residual spectral spatial attention network (RSSAN), which uses CBAM to search empty spectral features related to target pixel points and assigns different weights to them to obtain more discriminative features, thus improving the classification accuracy. Wang et al. [49] proposed a full-scale linked Unet network based on spatial–spectral joint perceptual attention for HS image and multispectral image fusion. Song et al. [50] proposed a cross-temporal interaction symmetric attention network for multitemporal HS images' change detection. In recent years, researchers have used the AM to study the interaction between HS images and LiDAR data. Mohla et al. [51] effectively used the self-AM to generate attention graphs to highlight their own spectral features. Meanwhile, they proposed cross-attention highlights modes between each other. Dong et al. [52] proposed a multibranch feature fusion network with self- and cross-guided attention (MB2FscgaNet), through the estimated spatial–spectral elevation features more effectively, making its produce more effective transmission. Wang et al. [53] proposed a novel dual attention fusion network (DAFNet). Liu et al. [54] proposed a central attention network (CAN) by combining HSI-tailored dense connections of the features of the hidden layers and the spectral information of the query pixel.

III. Methodology

A. Overall Architecture

This article proposes a DSHFNet based on multiattention for HS image and LiDAR classification. The overall framework is depicted in Fig. 1. The proposed approach incorporates a conventional CNN architecture as the primary framework and integrates a DSFE module to overcome the constraints of singular scale feature extraction. Moreover, the method uses multiple attention techniques to hierarchically fuse the extracted features, leading to improved fusion and ultimately better classification accuracy.

B. DSFE Module

During feature extraction, using an assessment of scale to identify the most appropriate scale for a specific object, the insufficiency of feature extraction encountered in conventional techniques can be overcome. Using the scale feature fusion module, small-scale features can be fused into large-scale features, which solves the problem of weight allocation among multiple scales.

DSFE first uses the feature map of minimum scale to train the subnetwork and obtains the similarity relationship between each pixel in the scale space, which is further judged. If the similarity reaches the specified standard, the feature graph under this scale is considered to have a good representation form. If the current subnetwork fails to reach the given standard, the subnetwork with a higher scale will be applied. At the same time, coarse features from the previous subnetwork are reused and merged into the current subnetwork. DSFE avoids unnecessary convolution of features at high spatial scales when samples can be accurately predicted by low spatial scale representations, thus reducing the computational budget and improving the computational efficiency.

DSFE comprises three blocks: Conv, scale feature discriminator, and scale feature fusion module. Within the Conv block, 2-D convolution with a kernel size of 3×3, batch normalization, and ReLU activation operations are used. The number of convolution cores is empirically set at 32 and 64. The detailed implementation for the Conv blocks is shown in Fig. 2. The scale feature discriminator calculates the similarity of samples at the scale and assigns appropriate computing

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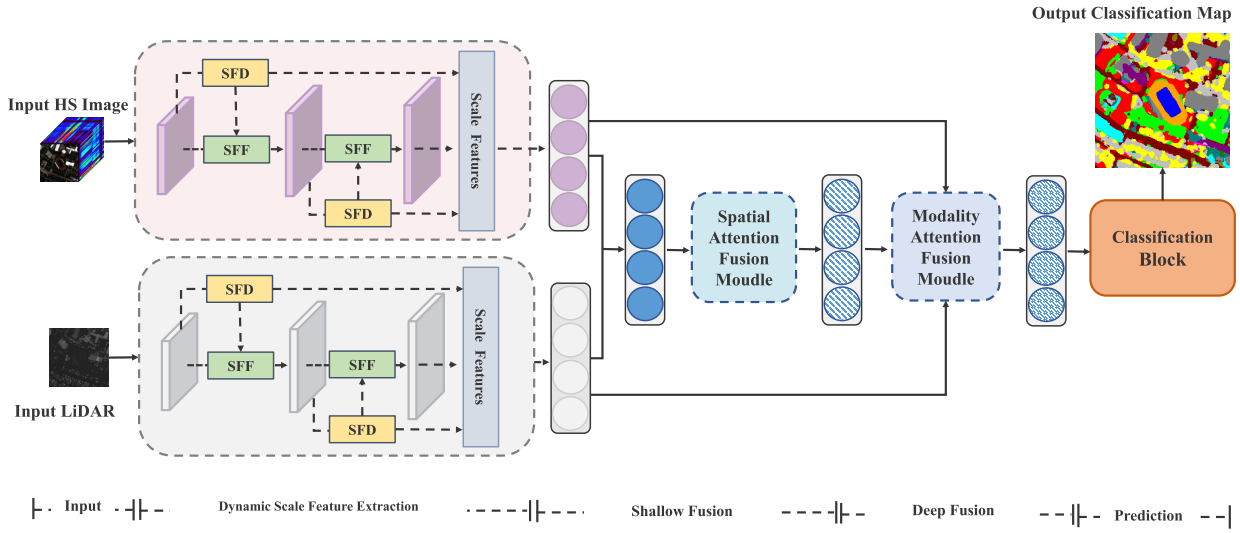

Fig. 1. Proposed DSHFNet framework for HS image and LiDAR fusion classification. First, a DSFE module is used to extract HS image and LiDAR features. Second, shallow fusion and extraction of joint features through spatial attention fusion module. Finally, the modality attention fusion module interacts the joint features with the features of the two modes to obtain complete features. The classifier is then used to obtain the classification results.

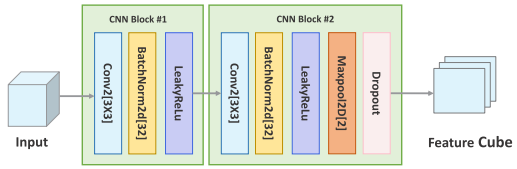

Fig. 2. Detailed implementation for the Conv blocks.

resources. Once the first output meets certain criteria, it exits the network. If the sample does not meet the standard, the low spatial scale features are fused into higher spatial scales through the scale feature fusion module. Various scenarios of DSFE are shown in Fig. 3.

The specific process of the scale feature discriminator is as follows: First, the features obtained through the Conv block are subjected to averaging and maximization, resulting in spatial information

$$F^{\text{avg}} = \frac{1}{C}\sum_{c=1}^{C} F_i \tag{1}$$

$$F^{\max} = \max(F_i) \tag{2}$$

where F_i is the feature of the input HS image and LiDAR patch obtained by the convolution operation. After that, the two outputs are connected and fed into a new convolutional layer, which is calculated by the sigmoid activation function

$$F^n = \text{sigmoid}\left(\text{conv}\left(\text{cat}\left[F^{\text{avg}}, F^{\max}\right]\right)\right) \tag{3}$$

where sigmoid. represents the activation function, conv. represents the convolution, and cat. represents the connection function. The similarity relationship between each pixel in the scale space is determined by the discriminator. If the similarity reaches a specified threshold, the pixel is considered similar to the central pixel in the feature map of that scale. Conversely, if the pixel does not meet the threshold, it is considered an

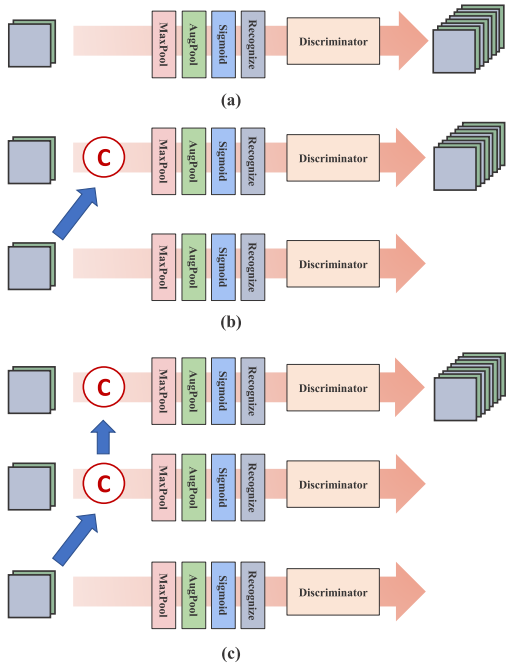

Fig. 3. Several cases of DSFE. (a) Subnetwork 1. (b) Subnetworks 1 and 2. (c) Subnetworks 1–3.

irrelevant pixel

$$f_{ij}^n = \begin{cases} f_{ij}^n, & f_{ij}^n > \beta \\ 0, & f_{ij}^n \le \beta \end{cases} \tag{4}$$

where the parameter β is used to determine the similarity between pixels in the respective scale space. Unrelated pixels are counted and compared with the total number of pixels in the current scale

$$\omega = \frac{\text{num}(0)}{\text{num}(\text{ALL})} \tag{5}$$

where num(0) is the number of pixels without correlation, and num(ALL) is the total number of pixels at this spatial

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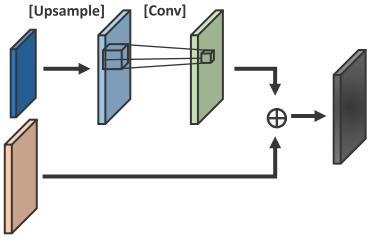

Fig. 4. Illustration of the scale feature fusion module.

scale. The smaller the ω, the greater the similarity in the scale space. If the similarity value is greater than the threshold σ, it signifies that the similarity conforms to the predetermined criterion. Then it is considered that the feature map at this scale has a good representation form and can effectively express the features and output the results, avoiding unnecessary convolution of features with high spatial scale, thus reducing the calculation budget and improving the calculation efficiency.

Conversely, if the similarity falls below the threshold of σ, it signifies that the similarity fails to meet the desired criteria and necessitates the use of a larger processing scale self-network to extract features from larger scale samples. To avoid the loss of feature information at small scales, the scale feature fusion module is used to integrate features from small scales into those of larger scales, as illustrated in Fig. 4.

First, the features at small scales are upsampled to the appropriate size. Convolutions were then applied to these features to obtain enhanced features at small scales

$$F_{\mathrm{SS}}{}' = \mathrm{conv}(\mathrm{upsample}(F_{\mathrm{SS}})) \tag{6}$$

where F_{SS} represents the original features at small scales, and upsample. refers to the upsampling operation. Next, a weighted fusion of small-scale and large-scale features is performed

$$F = F_{\mathrm{LS}} + (1-\omega)F_{\mathrm{SS}}{}' \tag{7}$$

where F_{LS} represents the original features at small scales.

If subnetwork 2 fails to produce dependable similarity, we proceed to repeat the same process for subnetwork 3. This iterative cycle continues until reliable similarities are obtained from a subnetwork or the final subnetwork is exhausted. The adaptive DSFE process generates HS images and LiDAR features which are then adaptively linearly weighted and mapped to the same spatial size using a fully connected layer to generate features F_H and F_L.

The above reasoning process is consistent with our intuition on land-covered classification. The "simple" category with representative features can be well distinguished, and pixels can be directly identified. The "hard" category with atypical features needs to be correctly identified according to the first law of geography based on the global information extracted from large-scale feature maps.

C. Hierarchical Fusion Based on Multiattention Module

1) Shallow Fusion Based on Spatial Attention Module (SF-SAM): The structure of the SF-SAM [48] is depicted in great detail in Fig. 5. These features are then fused using

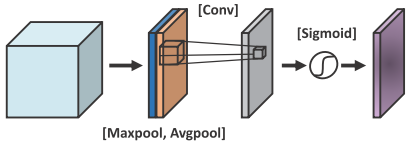

Fig. 5. Illustration of the spatial attention module.

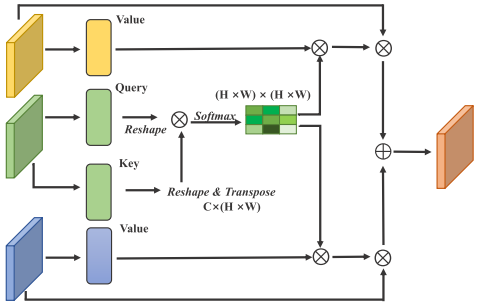

Fig. 6. Illustration of the modality attention module.

multimode fusion technology to generate the fusion feature F, which is calculated using the following equation:

$$F = \alpha F_H + (1-\alpha)F_L \tag{8}$$

where α is an adaptive weighting coefficient, which is learned and updated automatically by the network.

Given the significant differences between the two modalities, global average pooling and global max-pooling operations are then applied on the channel dimension of the feature map F to generate two distinct feature maps, namely, F_A and F_M. This approach enhances the spatial features through the AM, as it incorporates different processing methods and attributes. Overall, the aforementioned procedure can be summarized as follows:

$$F_A = \mathrm{Avgpooling}(F) \tag{9}$$

$$F_M = \mathrm{Maxpooling}(F). \tag{10}$$

Subsequently, F_A and F_M are merged to generate 2-D features. The resulting features are further processed through a 7×7 convolutional operation with a larger receptive field, followed by a sigmoid operation to fuse the features. This is then fed into a logistic regression model to obtain a low-dimensional feature F_{SJ}

$$F_{\mathrm{SJ}} = \mathrm{sigmoid}(\mathrm{conv}(\mathrm{cat}[F_A, F_M])). \tag{11}$$

Upon traversing the SF-SAM module, a shallow fusion approach is used to integrate multiple data sources, while simultaneously reducing the dimensionality of fusion features to alleviate network overhead.

2) Deep Fusion Based on Modality Attention Module (DF-MAM): The process of shallow feature fusion enhances the extraction of joint features through spatial attention modules. However, interactive capability between the two modalities is still lacking. We propose a modality attention module for deep fusion of multisource features, with its specific structure illustrated in Fig. 6.

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Process the joint features obtained from shallow fusion and features from two modalities through three convolutional layers to generate three new sets of features $\{Q_J, K_J, V_J\} \in \mathbb{R}^{C\times H\times W}$, $\{Q_H, K_H, V_H\} \in \mathbb{R}^{C\times H\times W}$, and $\{Q_L, K_L, V_L\} \in \mathbb{R}^{C\times H\times W}$. The calculation process of the deep fusion, which is based on the modality attention module, is defined as follows:

$$\mathrm{HMA}(Q_J,\ K_J,\ V_H) = \mathrm{Softmax}\left(Q_J {K_J}^T\right) V_H \tag{12}$$

$$\mathrm{LMA}(Q_J,\ K_J,\ V_L) = \mathrm{Softmax}\left(Q_J {K_J}^T\right) V_L \tag{13}$$

where T represents the transpose operation. Optimizing the aggregation of feature maps from different modalities using varying weights

$$F_{Hj} = F_H \otimes (\theta_1 \cdot \mathrm{HMA}) \tag{14}$$

$$F_{Lj} = F_L \otimes (\theta_2 \cdot \mathrm{LMA}) \tag{15}$$

where $\otimes$ represents the inner product operation of the matrix. The parameter θ is a dynamic weighting coefficient that is trained and updated autonomously through the network's learning process.

The final joint feature is obtained by calculating the sum of all the features after deep fusion

$$F_{\mathrm{DJ}} = F_{Hj} + F_{Lj}. \tag{16}$$

D. Classification Block

To generate the ultimate outcome, a CNN-based classifier is used, consisting of three blocks. The first block applies 2-D convolution with a 1×1 kernel size, followed by batch normalization and ReLU activation. This block comprises 32 convolution kernels. The second block uses adaptive global average pooling to reduce the spatial size to 1×1, capturing pixel category information. The third block consists of a fully connected layer that maps the number of spectral channels to the designated number of ground object categories j. Ultimately, the SoftMax classification layer is implemented to predict the pointwise classification probabilities. This approach avoids repetition and ensures a clear and concise description

$$\mathrm{pred}(i,\ j) = \frac{1}{\sum_{j=1}^{K} \exp(\theta_j | F_{\mathrm{DJ}})} \begin{bmatrix} \exp(\theta_1 | F_{\mathrm{DJ}}) \\ \cdots \\ \exp(\theta_K | F_{\mathrm{DJ}}) \end{bmatrix} \tag{17}$$

where θ_j represents the jth column of the weights of the prediction layer, K is the total number of the land-cover classes, and $\mathrm{pred}(i, j)$ is the class probabilities for current position.

E. Network Training

During the network training process, the proposed framework takes in the training set samples in the form of $\{(X_H^n, X_L^n), Y^n\}_{n=1}^U$, wherein (X_H^n, X_L^n) refers to the pair of pixels associated with the same pixel coordinates in the HS image and LiDAR in the training sample, and Y^n denotes the actual labels corresponding to these pairs. The total number of training samples is denoted by U. To address the issue of disparate quantities across different categories, we use the weighted cross-entropy loss as the underlying loss function

$$\mathcal{L}_{W-C} = -\frac{1}{U} \sum_{n=1}^{U} \sum_{j=1}^{K} \omega_j \log \Pr\left(Y^n = j | \left(X_H^n, X_L^n\right); \theta\right) \tag{18}$$

where ω_j is the jth weight, and $\Pr(Y^n = j|(X_H^n, X_L^n); \theta)$ is the probability that the pixel belongs to the jth class. For weights, the definition is as follows:

$$\omega_j = \frac{n_j}{U} \tag{19}$$

where n_j is the number of class j in the ground-truth training sample. To achieve maximal parameters, we use an Adam optimizer with Nesterov momentum [55]. Algorithm 1 gives the overall network training process of DSHFNet.

Algorithm 1 Tasks Completed by the Algorithm

Require: Hyperspectral image, H; LiDAR-based DSM, L; training label, Y; training epochs, $epochs$; patch size, r;
Ensure: Classification map, P;

1: Initialize all weights;
2: The input block corresponding to H and L is obtained according to r, $2r$, and $3r$;
3: **while** $epoch < epochs$ **do:**
step 1: Extract shallow HS features F_H of patch size r from Conv blocks;
step 2: Calculate similarity values ω by (1)–(5);
step 3: Compare ω with threshold σ
if $\omega > \sigma$ or patch size $= 3r$:
break;
else
Extract shallow features HS F_H of patch size $2r$ from Conv blocks, then repeat step 2;
step 4: Extract LiDAR features, steps consistent with steps 1–3;
step 5: Extract joint features F_{SJ} by (8)–(11);
4: step 6: Extract joint features F_{DJ} by (12)–(16);
5: step 7: Train the network to obtain the highest weight result;
6: Use classification function $SoftMax$ according to W to obtain P.

IV. Experiments and Analysis

A. Dataset Description

To validate the efficacy of the proposed algorithm, three publicly available multisensor RS image classification datasets are used as experimental datasets, including the Houston2013 dataset, Trento dataset, and Augsburg dataset. Detailed parameters are shown in Table I.

B. Parameter Tuning

In this article, we implement our programs with Python 3.7, and the convolutional neural network part is implemented using PyTorch. All the experiments were conducted on a PC configured with Intel Core i7-10870 h, GeForce RTX 2070 with Max-Q Design, and Windows 10.

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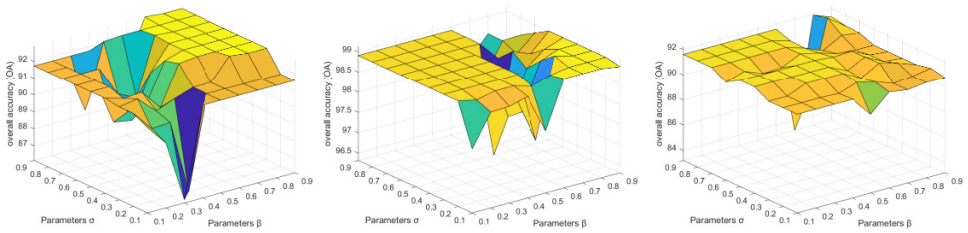
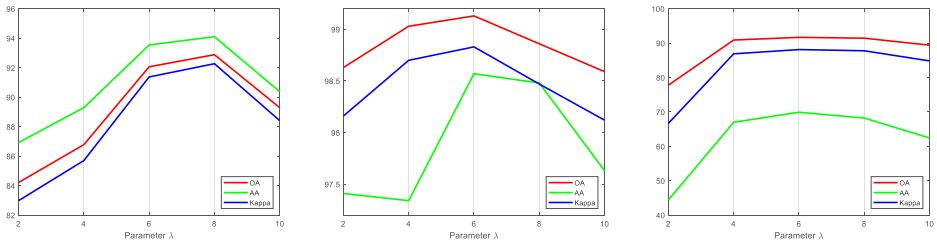


(a) (b) (c)

Fig. 7. Comparison of model performance across varying parameters. (a) Houston2013. (b) Trento. (c) Augsburg.



(a) (b) (c)

Fig. 8. Comparison of model performance across varying input patch sizes. (a) Houston2013. (b) Trento. (c) Augsburg.

TABLE I
DATASET DESCRIPTION

Dataset	Houston2013 [56]		Trento [21]		Augsburg [1]	
Location	Houston, Texas, USA		Trento, Italy		Augsburg, Germany	
Sensor Type	HSI	LiDAR	HSI	LiDAR	HSI	LiDAR
Image Size	349×1905	349×1905	600×166	600×166	332×485	332×485
Spatial Resolution	2.5m	2.5m	1m	1m	30m	30m
Number of Bands	144	1	63	1	180	1
Wavelength Range	0.38-1.05m	/	0.42-0.99m	/	0.4-2.5m	/
Sensor Name	CASI-1500	/	AISA Eagle	Optech ALTM 3100EA	HySpex	DLR-3 K

The performance of a learning algorithm model is determined by the design of network structure, in this section, we discuss the model performance by varying the hyperparameters of the network, including weight parameters σ and β of the dynamic scale feature discriminator, multiscale input patch size, and network learning rate. Meanwhile, we use overall accuracy (OA), average accuracy (AA), and Kappa coefficient as the evaluation criterions to compare the classification results with the ground-truth maps, where a higher value indicates a better classification performance.

1) Dynamic Scale Feature Discriminator Parameter Comparison: In Section III-B, the DSFE stage uses pixel similarity within a spatial scale to determine the appropriate input scale, with weight parameters σ and β influencing the module. We varied their values and compared the classification results across three datasets. Fig. 7 presents the OA as we manipulated the weight parameters on these datasets. It can be observed that when a certain parameter is too large or too small, the experimental results tend to show the same numerical values, indicating that under this set of parameters, the dynamic adaptive feature extraction module did not perform optimally. To ensure the effectiveness of the module, we need to establish a balance between σ and β to achieve the optimal effect. We set these weight parameters to $\sigma = 0.3$ and $\beta = 0.7$ in all three datasets used in our experiments.

2) Multiscale Input Patch Size Comparison: The decision on the appropriate patch size is paramount in image classification as using a large patch size introduces unnecessary information while using a small patch size can make it difficult to discern important information. For multiscale feature input, patch size becomes even more important, and different patch sizes are applied to yield different imaging results. Fig. 8 illustrates the impact of input patch sizes λ ranging from 2 to 10. The results show that 8 is the optimal parameter for the Houston2013 dataset, while 6 is optimal for the other two datasets. This variation is attributed to the differing spatial scale sizes across the various datasets.

3) Learning Rate Comparison: Learning rate plays a critical role in the deep learning process as a vital hyperparameter. The primary objective is to regulate the network's mass by computing and applying the gradient of the loss function. If learning rate is too high, it may result in a stride in parameter updating that surpasses the optimal value. This, in turn, leads to fluctuations in the loss function value during parameter optimization and ultimately results in network convergence failure. On the contrary, a learning rate that is too low may result in the network parameters being trapped in a high local minimum during parameter search, failing to explore better local minimums that are more likely to be global. Fig. 9 showcases the classification performance with learning rates ranging from 0.005 to 0.00005. The results show that 0.0005 is the optimal parameter for three datasets.

C. Comparison and Analysis of Classification Performance

To showcase the effectiveness of the proposed approach, this section compares it with SOTA algorithms, including: TBCNN [13], multimodal deep learning framework (MDL-RS) [57], deep encoder–decoder network (EndNet) [21], cross-channel reconstructed convolutional neural network

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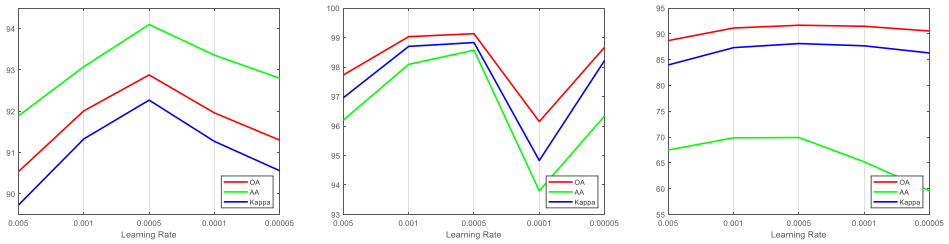

Fig. 9. Comparison of model performance across varying learning rates. (a) Houston2013. (b) Trento. (c) Augsburg.

TABLE II
CLASSIFICATION ACCURACIES' COMPARISON WITH THE HOUSTON2013 DATASET

No.	Class(Train/Test)	TBCNN	MDL-RS	EndNet	CCR-Net	Glt-Net	MAHiDFNet	CALC	HRWN	ExViT	DHViT	Proposed
1	Health grass (198/1053)	83.10	83.10	81.58	83.00	83.10	83.10	82.24	85.77	91.55	81.01	**97.44**
2	Stressed grass (190/1064)	84.10	81.58	83.65	84.87	84.17	82.61	83.93	**86.52**	85.15	85.15	85.15
3	Synthetic grass(192/505)	**100.00**	**100.00**	**100.00**	**100.00**	93.07	**100.00**	93.47	98.42	98.61	93.47	97.62
4	Trees(188/1056)	93.03	99.72	93.09	92.14	93.38	93.37	98.86	93.41	98.61	80.40	**100.00**
5	Soil (186/1056)	**100.00**	99.81	99.91	99.81	99.81	99.72	99.72	100.00	**100.00**	99.53	**100.00**
6	Water (182/143)	99.30	95.10	95.10	95.80	**100.00**	**100.00**	98.60	90.77	98.60	95.80	**100.00**
7	Residential (196/1072)	92.82	90.02	82.65	**95.34**	93.66	92.26	90.21	94.95	88.90	92.16	88.34
8	Commercial (191/1053)	82.34	87.94	81.29	81.39	90.60	91.55	81.58	93.01	**93.35**	92.21	82.15
9	Road (193/1059)	84.70	81.59	88.29	84.14	**92.16**	86.02	84.99	87.94	89.52	81.78	89.61
10	Highway (191/1036)	65.44	86.68	89.00	63.22	67.18	66.31	68.15	69.85	65.54	68.73	**99.32**
11	Railway (181/1054)	88.24	89.37	83.78	90.32	83.78	**95.92**	95.16	94.82	95.83	79.60	80.17
12	Parking lot 1 (192/1041)	89.53	85.69	90.39	93.08	96.54	94.24	93.56	92.54	90.39	94.81	**98.75**
13	Parking lot 2 (184/285)	92.28	83.16	82.46	88.42	91.23	**98.60**	87.37	95.52	90.53	88.42	92.98
14	Tennis court (181/247)	96.76	**100.00**	**100.00**	96.36	97.98	**100.00**	99.60	**100.00**	99.60	**100.00**	**100.00**
15	Running track (187/473)	99.79	98.73	98.10	99.37	**100.00**	**100.00**	99.58	**100.00**	89.85	71.25	**100.00**
	OA%(2832/12197)	89.98	89.60	88.52	88.15	89.50	90.06	88.91	91.22	90.62	85.82	**92.88**
	AA%	90.11	90.83	89.98	89.89	91.11	92.25	90.47	92.23	91.76	86.95	**94.10**
	Kappa×100	86.98	88.75	87.59	87.19	88.64	89.24	87.98	90.52	89.83	84.61	**92.27**

(CCR-Net) [17], Glt-Net [28], MAHiDFNet [41], coupled adversarial learning-based classification (CALC) [23], HRWN [38], ExViT [27], and DHViT [42]. All the comparative experiments used HS-LiDAR data. All the methods take multimodal data as input. TBCNN, MDL-RS, CCR-Net, MAHiDFNet, and HRWN use CNNs as the backbone network, EndNet uses SAEs as the backbone network, CALC uses GANs as the backbone network, ExViT and DHViT use Transformers as the backbone network, and Glt-Net uses a combination of CNNs and Transformers as the backbone network. Among them, Glt-Net extracted features at multiscales, MAHiDFNet, HRWN, Glt-Net, and DHViT are the hierarchical network, and MAHiDFNet, Glt-Net, ExViT, and DHViT used attention module. The network is entered in the same way as its exposed code. To fairly compare the advantages and disadvantages among the methods, training iterations and learning rates are exactly the same.

Tables II–IV display the classification accuracies comparisons of the three datasets, wherein our proposed method demonstrates significantly improved results. For the Houston2013 dataset, the OA reaches 92.88%. The accuracy of most categories is above 85%, and the categories of trees, soil, water, tennis court, and running track are 100%. Dynamic scale fully extracts the information under different land cover distributions and maximizes the use of the consistency of the surrounding land cover, which can effectively avoid the appearance of singular pixels, and completely retains the characteristics of refined pixels to ensure that they are not merged by the surrounding large categories. By comparing the objective evaluation results, it can be seen that compared with other hierarchical networks, the proposed hierarchical fusion strategy can make the fusion of multisource data more comprehensive, and all the objective evaluation results are better than other methods, especially Kappa has a great improvement. This is because the hierarchical fusion strategies of other methods often only consider the connection between shallow and deep features and do not effectively analyze the unique features and joint features of the modes together. The proposed method takes into account the superficial texture features and the deep semantic features, and at the same time uses the unique features and effectively integrates the joint features. For example, when distinguishing between three types of grass, both achieved the best classification results. The application of AM also makes the extraction of joint features and the fusion of deep features more reliable. For instance, in the highway category, DSHFNet achieved significant improvements, further demonstrating effectiveness of hierarchical fusion and AM.

In addition to the accuracy of quantitative classification, we also validate the effectiveness from a visual perspective. The classification maps are presented in Figs. 10–12. It is apparent that the powerful nonlinear fitting ability of deep learning can fully excavate deep semantic features, and most of the obtained classification maps have very smooth boundaries. However, since the patch is used as the input method in all the deep learning methods, scattered land covered in small categories is often wrongly classified into other large categories

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TABLE III

CLASSIFICATION ACCURACIES' COMPARISON WITH THE TRENTO DATASET

No.	Class(Train/Test)	TBCNN	MDL-RS	EndNet	CCR-Net	Glt-Net	MAHiDFNet	CALC	HRWN	ExViT	DHViT	Proposed
1	Apple trees (129/3905)	98.07	88.58	88.19	**100.00**	98.87	98.90	97.26	99.97	99.56	98.36	99.49
2	Building (125/2778)	95.21	95.86	98.49	98.88	99.14	98.70	**100.00**	**100.00**	98.13	99.06	98.74
3	Ground (105/374)	93.32	93.58	95.19	79.68	99.47	98.93	89.57	82.09	76.47	67.65	**99.73**
4	Woods (154/8969)	99.93	99.22	99.30	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
5	Vineyard (184/10317)	98.78	83.82	91.96	94.79	99.68	**100.00**	99.75	**100.00**	99.93	98.89	**100.00**
6	Roads (122/3052)	89.98	76.51	90.14	88.07	90.23	93.41	87.45	93.41	**93.84**	87.98	93.45
	OA%(819/29395)	97.92	90.65	94.17	96.57	98.67	99.03	98.16	99.08	98.80	98.00	**99.13**
	AA%	96.19	89.60	93.88	93.57	97.96	98.32	95.68	95.91	94.66	92.16	**98.57**
	Kappa×100	96.81	86.28	92.22	95.43	98.22	98.70	97.48	98.77	98.39	97.31	**98.83**

TABLE IV

CLASSIFICATION ACCURACIES' COMPARISON WITH THE AUGSBURG DATASET

No.	Class(Train/Test)	TBCNN	MDL-RS	EndNet	CCR-Net	Glt-Net	MAHiDFNet	CALC	HRWN	ExViT	DHViT	Proposed
1	Forest(146/13361)	95.29	84.97	89.70	93.47	90.82	93.98	94.16	96.63	91.83	90.45	**97.60**
2	Residential-Area(264/30065)	96.51	91.56	86.75	**96.86**	94.96	95.41	95.32	95.44	95.38	90.87	92.94
3	Industrial-Area(21/3830)	68.59	8.17	24.26	82.56	71.44	84.31	86.18	52.32	43.32	61.20	**87.13**
4	Low-Plants(248/36609)	82.43	78.07	76.77	84.45	92.25	92.69	95.57	86.88	91.13	82.82	**96.38**
5	Allotment(52/523)	**75.52**	26.00	34.42	44.36	69.41	63.67	0.00	54.30	41.11	21.80	64.05
6	Commercial-Area(7/1638)	15.99	2.14	9.46	0.00	7.39	7.33	6.05	5.31	**26.01**	23.50	2.50
7	Water(23/1507)	48.37	36.30	46.78	40.48	43.33	51.69	**55.96**	43.80	42.07	7.43	48.77
	OA%(761/77533)	87.31	79.11	78.25	87.82	89.13	90.76	91.46	87.40	87.82	83.06	**91.67**
	AA%	68.96	46.74	52.29	63.17	67.08	69.69	61.89	62.10	61.41	54.01	**69.91**
	Kappa×100	82.05	67.52	68.11	82.48	84.51	86.92	88.04	81.78	82.44	75.91	**88.12**

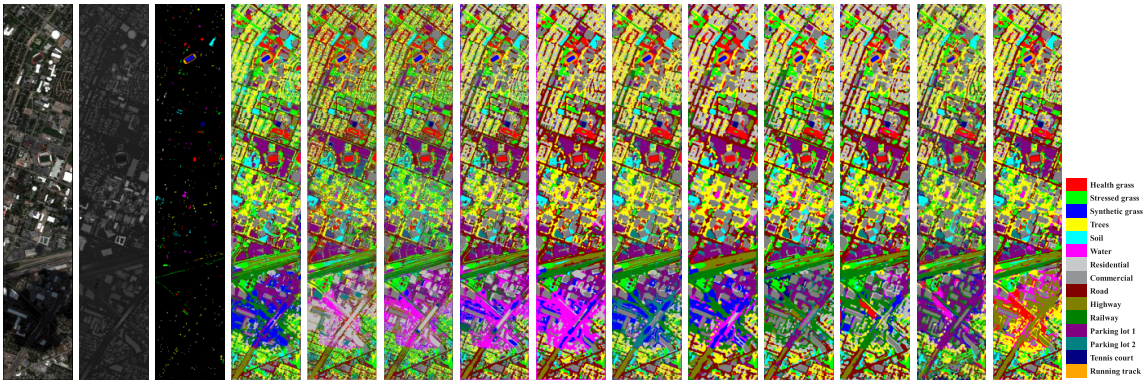

Fig. 10. Classification maps generated by different comparison methods for the Houston2013 dataset. (a) Pseudocolor image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN (89.98%). (e) MDL-RS (89.60%). (f) EndNet (88.52%). (g) CCR-Net (88.15%). (h) Glt-Net (89.50%). (i) MAHiDFNet (90.06%). (j) CALC (88.91%). (k) HRWN (91.22%). (l) ExViT (90.62%). (m) DHViT (85.82%). (n) Proposed (92.88%).

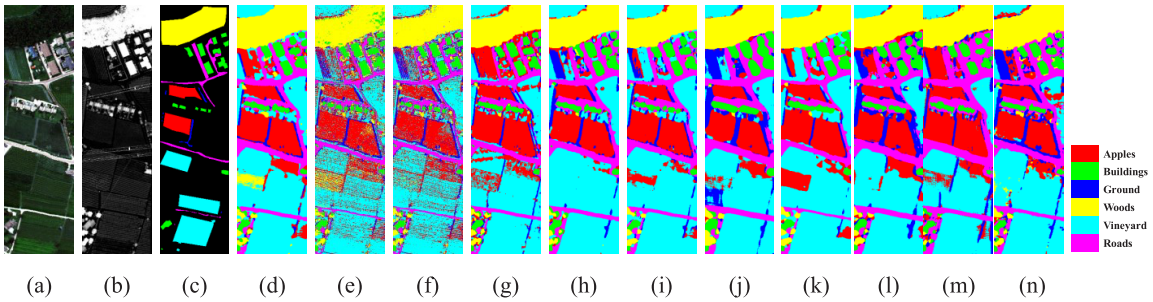

Fig. 11. Classification maps generated by different comparison methods for the Trento dataset. (a) Pseudocolor image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN (97.92%). (e) MDL-RS (90.65%). (f) EndNet (94.17%). (g) CCR-Net (96.57%). (h) Glt-Net (98.67%). (i) MAHiDFNet (99.03%). (j) CALC (98.16%). (k) HRWN (99.08%). (l) ExViT (98.80%). (m) DHViT (98.00%). (n) Proposed (99.13%).

around them. Our proposed algorithm can more effectively and accurately distinguish small class features by dynamic scale, so as to obtain ideal classification maps, textures and edge details, especially by presenting more realistic and finer details. Meanwhile, as can be seen from the right part of the Houton2013 dataset, EndNet, CCR-Net, and Glt-Net all classify this area as Category 6, water. This is because the HS image of this scene has a relatively serious cloud phenomenon, which has a serious coverage effect on the land covered. The fusion effect of HS image and LiDAR of these methods is not

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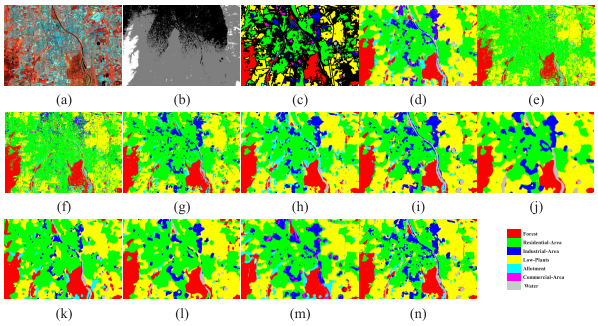

Fig. 12. Classification maps generated by different comparison methods for the Augsburg dataset. (a) Pseudocolor image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN (87.31%). (e) MDL-RS (79.11%). (f) EndNet (78.25%). (g) CCR-Net (87.82%). (h) Glt-Net (89.13%). (i) MAHiDFNet (90.76%). (j) CALC (91.46%). (k) HRWN (87.40%). (l) ExViT (87.82%). (m) DHViT (83.06%). (n) Proposed (91.67%).

good, so the classification results are more dependent on the classification results of HS image, which leads to such wrong results. Our proposed method can more effectively integrate features through hierarchical fusion and AM and better classify cloud and fog-covered image areas using the penetration effect of LiDAR. In addition, there is a highway in this area that is not adequately captured by most methods, but our proposed method presents it more accurately. The objective evaluation results of the experiments also demonstrate the clear advantage of our method in the highway class compared with other methods. This is because the DSFE is better suited for precise extraction of specific land cover types. While achieving excellent results in the category of highways, due to the dynamic scale, there may be minor errors in the allocation of feature weights for certain land cover types in the hierarchical fusion. As can be seen in the visualization results of our proposed algorithm, some land cover pixels in the lower right corner were also misclassified as highways. Taking into account both the objective evaluation results and visual images, we have chosen to present the most suitable results in this article.

Currently, most of the deep-learning-based methods that achieve high classification accuracy rely on a significant quantity of labeled data. However, obtaining these labeled samples for RS data can be time-consuming and challenging, and sometimes even impossible to complete. Efficiently extracting accurate features regarding small sample sizes is a significant challenge and worth exploring for multisource RS fusion classification. To evaluate the performance of this algorithm under different numbers of training samples and its robustness under small sample sizes, we performed a variety of experiments and analyzed the findings. As shown in Fig. 13, it can be seen that in a few cases, HRWN achieves more results, and in other cases, our proposed method achieves the best results. Notably, the proposed method maintained good performance using only 20% of the training data, exceeding some methods that used a larger percentage of training data. This is particularly advantageous in multisource RS image classification task.

D. Ablation Study

To substantiate the efficacy of our novel modules, namely, the DSFE module, the hierarchical fusion strategy, and the AM module, we conducted ablation studies for each of them. The findings from the ablation experiments further authenticate the potency of the envisaged breakthroughs.

TABLE V
OBJECTIVE EVALUATION INDICATORS OF DIFFERENT HIERARCHICAL FUSION STRATEGIES FOR THE HOUSTON2013 DATASET

	MA2CNN	DHViT	MAHiDFNet	HRWN	PtoPCNN	Proposed
OA	85.54	85.82	90.06	91.22	92.48	**92.88**
AA	87.19	86.95	92.25	92.23	93.55	**94.10**
Kappa	84.36	84.61	89.24	90.52	91.87	**92.27**

1) Effect of DSFE Module: The DSFE module is an important component in the proposed network. The influence of various sizes of characteristics on the accuracy of classification is illustrated in Fig. 14. Compared with any single-scale feature, the fusion of multiscale features produces superior classification results. This suggests that multiscale features can effectively combine different scale features to obtain more information. Furthermore, the results of DSFE exceed those of using multiscale features as inputs, especially for the AA method. This indicates that the DSFE module selects the most suitable scale feature as the input feature by weighted fusion of different scale features, considering both the comprehensiveness of multiscale features and avoiding redundancy of multiscale features, and plays a positive and effective role.

2) Effect of Hierarchical Fusion: Hierarchical fusion is an important innovation in the proposed framework. Fig. 15 shows the ablation results of different fusion methods. It is apparent that shallow fusion pays more attention to single-source information such as texture and structure of features, while deep fusion pays more attention to semantic information of features, which is more likely to appear adjacent approximate features annexation phenomenon, especially in AA of the Trento dataset. Hierarchical fusion makes the feature fusion more adequate through the fusion of two different positions and avoids the problem of unipolar fusion overhighlighting shallow or deep features, which plays a good role.

To better demonstrate the advantages of the proposed hierarchical fusion strategy, we conducted a comparison experiment between the proposed algorithm and some SOTA hierarchical fusion algorithms on the dataset of Houston2013, including multimodal attention-aware convolutional neural networks (MA2CNNs) [58], DHViT [42], MAHiDFNet [41], HRWN [38], and PtoPCNN [39]. The results are shown in Table V. It can be seen that the proposed method achieves optimal results, especially on the evaluation index AA. This is because different from other methods, our proposed method makes full use of the joint features of different modes and takes them as part of the fusion, so as to make the most efficient use of the common features of multisource data and get the best results.

3) Effect of AM: The quality of fusion results heavily relies on AM, making it the most imperative component in the fusion process. Fig. 16 shows an ablation study of AM. The AM also plays an interactive role in the data of different sensors, which makes more effective use of the complementarity between different sensor data and plays a positive role, especially in the Houston2013 dataset. This is because the dataset is an

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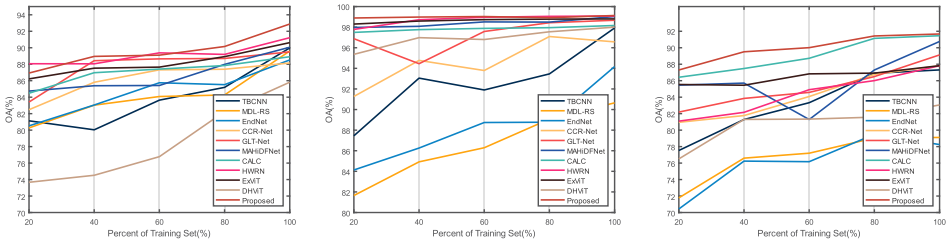

Fig. 13. Evaluating classification performance across varying sizes of training samples. (a) Houston2013. (b) Trento. (c) Augsburg.

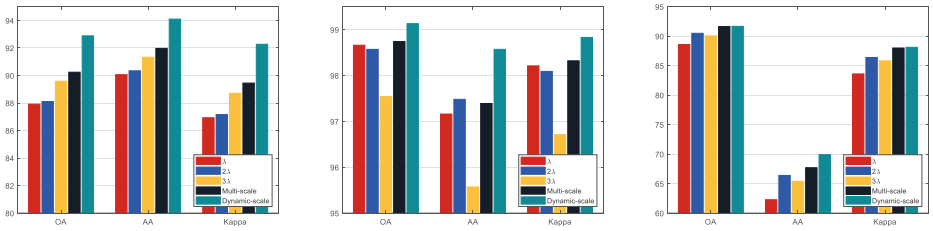

Fig. 14. Effects of DSFE module on classification effectiveness. (a) Houston2013. (b) Trento. (c) Augsburg.

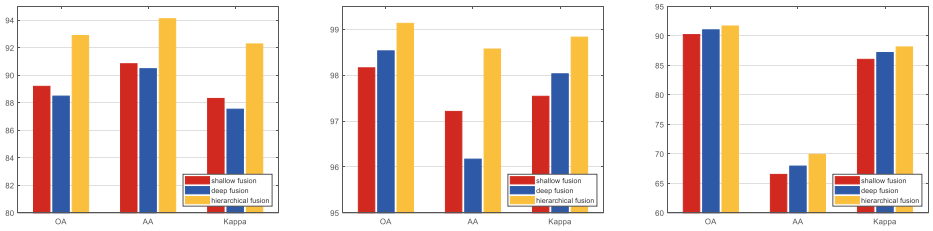

Fig. 15. Effects of hierarchical fusion on classification effectiveness. (a) Houston2013. (b) Trento. (c) Augsburg.

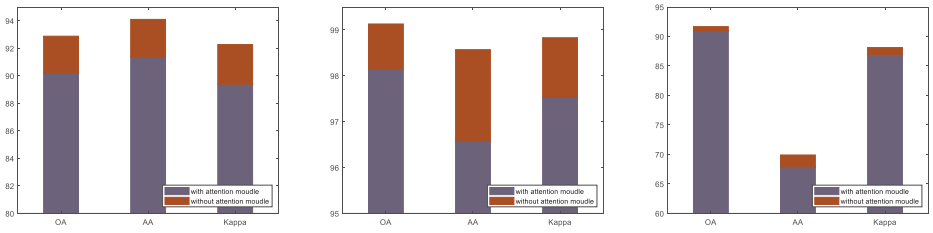

Fig. 16. Effects of AM on classification effectiveness. (a) Houston2013. (b) Trento. (c) Augsburg.

urban area with poor continuity of land covered, and the disadvantage can be effectively made up through AM. On the Augsburg dataset, the significant variation in the number of samples among each individual sample makes it difficult for AM to accurately discern the surroundings, resulting in the smallest incremental improvement.

V. Conclusion

HS images and LiDAR fusion for LULC classification are becoming a hotspot, but previous methods have limitations in feature extraction. In this article, to address this, the DSHFNet is proposed for fusion and classification. DSHFNet uses DSFE to effectively use features while avoiding the high dimensionality of multiscale features. Different attention modules are used for hierarchical fusion, achieving complete feature complementarity. Through experimentation, it has been demonstrated that DSHFNet outperforms the existing methods. The strategies applied, such as the DSFE module and hierarchical fusion approach, provide valuable insights for multimodal data fusion in various application domains. This can also serve as a reference for researchers in the fields of RS image processing. Furthermore, we have noted that the

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research of RS image fusion and classification based on small samples has received more and more attention in recent years. These aspects deserve more attention, and we will study them in future research work.

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Yining Feng received the M.Eng. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Geography, Liaoning Normal University.

His research interests include multimodal remote sensing image processing.

Liyang Song received the B.Sc. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2022. She is currently pursuing the M.Sc. degree with the School of Computer Science and Artificial Intelligence, Liaoning Normal University.

Her research interests include multimodal remote sensing image classification.

Lu Wang received the B.Eng. degree from the School of Information Engineering, Dalian Ocean University, Dalian, China, in 2021. She is currently pursuing the M.Sc. degree with the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian.

Her research interests include semi-supervised remote sensing image classification.

Xianghai Wang received the Ph.D. degree from Jilin University, Changchun, China, in 1999.

He is currently a Professor with the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian, China. His research interests include image processing, multimedia information processing, and remote sensing image processing.

Dr. Wang is a Senior Member of the China Computer Federation (CCF).

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